

# Study on Establishing Integrated Open Access Research Activity Management System

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## 1. Introduction

Scholarly communication and research activities are part of a cyclical process involving the discovery and analysis of research data/papers, paper writing, submission and peer review, publication, and the subsequent use of publications in other research endeavors. This process primarily involves scholarly articles. However, access restrictions owing to high subscription fees for academic journals hinder the entire research process. Unrestricted access to scholarly papers is essential for fostering an open research environment, leading to the emergence of the concept of open access [1].

In scholarly communication, researchers play multiple roles such as authors and peer reviewers. These roles are crucial aspects of scholarly communication but are complex and time-consuming tasks. To facilitate public access to research papers and encourage the openness of more articles, it is imperative to support researchers in efficiently performing their roles in an open-access environment. In this context, this study focuses on the research activity management system My Activity, which aims to enhance the efficiency of managing the entire research activity on the open-access platform AccessON [2].

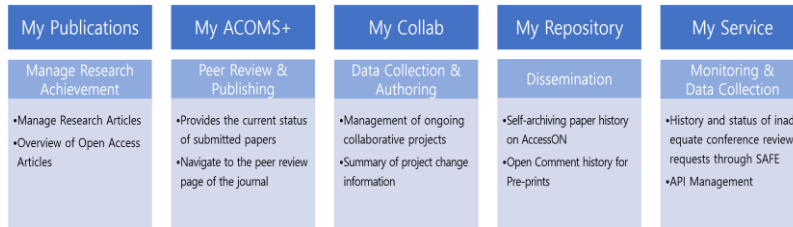


Figure 1. Scholarly communication and my activity

## 2. Structure of my activity

The integrated research activity management service in AccessON, known as My Activity, is designed to assist researchers in tracking the progress of every research activity within the AccessON integrated system, including ACOMS+ and the AccessON Peer Review Management System. This serves as a valuable tool to guide researchers in their next steps more conveniently and offers a convenient list of open-access articles for easy management. My Activity contains five menus, each tailored to support a specific step in the scholarly communication process.

My Activity relies on seamless data integration with related systems and the AccessON platform to provide accurate information about research articles and their current status. Notably, it utilizes ORCID [3], a unique and persistent researcher identifier, to accurately identify work and share the peer review history of ACOMS+ with ORCID.



Figure 2. Data integration and outcomes

## 3. Establishment of open access article management system in AccessON

My Publications aims to provide researchers with a comprehensive list of open-access articles. Given the diverse avenues of open access, including the self-archiving 'green' route and open-access publishing via the 'gold' route, and the multitude of journal options available for submissions, compiling and summarizing a researcher's open-access publications can be complex. Researchers often require effective management of their articles for various purposes, including evaluation. My Publications streamlines this process by gathering article data from Discover and Self-Archiving through the following steps.

Among the articles in the AccessON database, the author's article, identified by his or her ORCID, is sent to My Publications. In addition, a self-archived open article in AccessON is added to My Publications to collect Green Open Access Article data.



Figure 3. My publications process

#### 4. My activity for submission and peer review

One of the most characteristic parts of My Activity is My ACOMS+, which provides information about the tasks conducted in ACOMS+, an integrated AccessON system. In scholarly communication, researchers perform multiple roles, such as peer reviewers, EIC, and editors. In addition, they have performed these roles in multiple journals. This makes researchers look at the different submission and peer review pages of each journal to determine the status of their submissions or peer reviews, which wastes time.

My ACOMS+ resolves this challenge by consolidating the task information across multiple journals on a single page. This allows researchers to efficiently check the status of submissions and peer reviews across different journals by providing direct links to specific submissions and peer review pages. This functionality was made possible by the integrated member systems AccessON and ACOMS+. Because both ACOMS+ and AccessON are seamlessly integrated, user accounts are unified, and data are shared between the two systems. Consequently, updates or changes in ACOMS+ are instantly reflected in My ACOMS+. My ACOMS+ comprises sections such as My Journals, My Submissions, and My Peer Reviews. Notably, when the peer review process of an open peer review journal or hybrid journal is concluded, My Peer Review in AccessON facilitates the seamless transfer of this information to the ORCID, serving as a comprehensive record of the researchers' contributions.

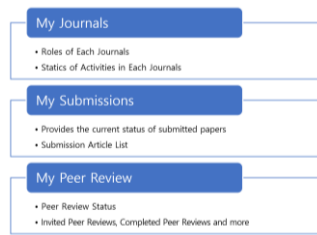


Figure 4. My ACOMS+ composition

| No. | Journal                                            | Author | Reviewer | Editor | EIC |
|-----|----------------------------------------------------|--------|----------|--------|-----|
| 1   | Journal of Christian Education in Korea            | ✓      | ✓        |        |     |
| 2   | Knowledge Management Research                      | ✓      |          |        |     |
| 3   | Test Journal                                       | ✓      | ✓        |        |     |
| 4   | test journal                                       | ✓      | ✓        |        |     |
| 5   | JOURNAL OF INFORMATION SCIENCE THEORY AND PRACTICE | ✓      |          |        |     |

Figure 5. My Journals Example

#### 5. My activity for co-authoring and self-archiving

As part of the service for the entire research activity, My Collab and My Repository support authors' co-authoring and self-archiving. Both provide real-time alarm services for researchers to understand what happens in ongoing or archived articles. In particular, My Repository provides a self-archiving article recommendation service using ORCID. Based on the ORCID linked to the AccessON account and comparison with self-archiving policy sites, when the article's embargo ends, it is listed on the My Repository page.

#### 6. Conclusion

To achieve open access, researcher participation is essential, and My Activity is the first step toward making it more efficient and systemic. The continuous improvement of My Activity reflecting users' opinions will make it more convenient, which could result in greater participation of researchers in the green and gold routes.

#### Acknowledgments

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#### 7. References

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